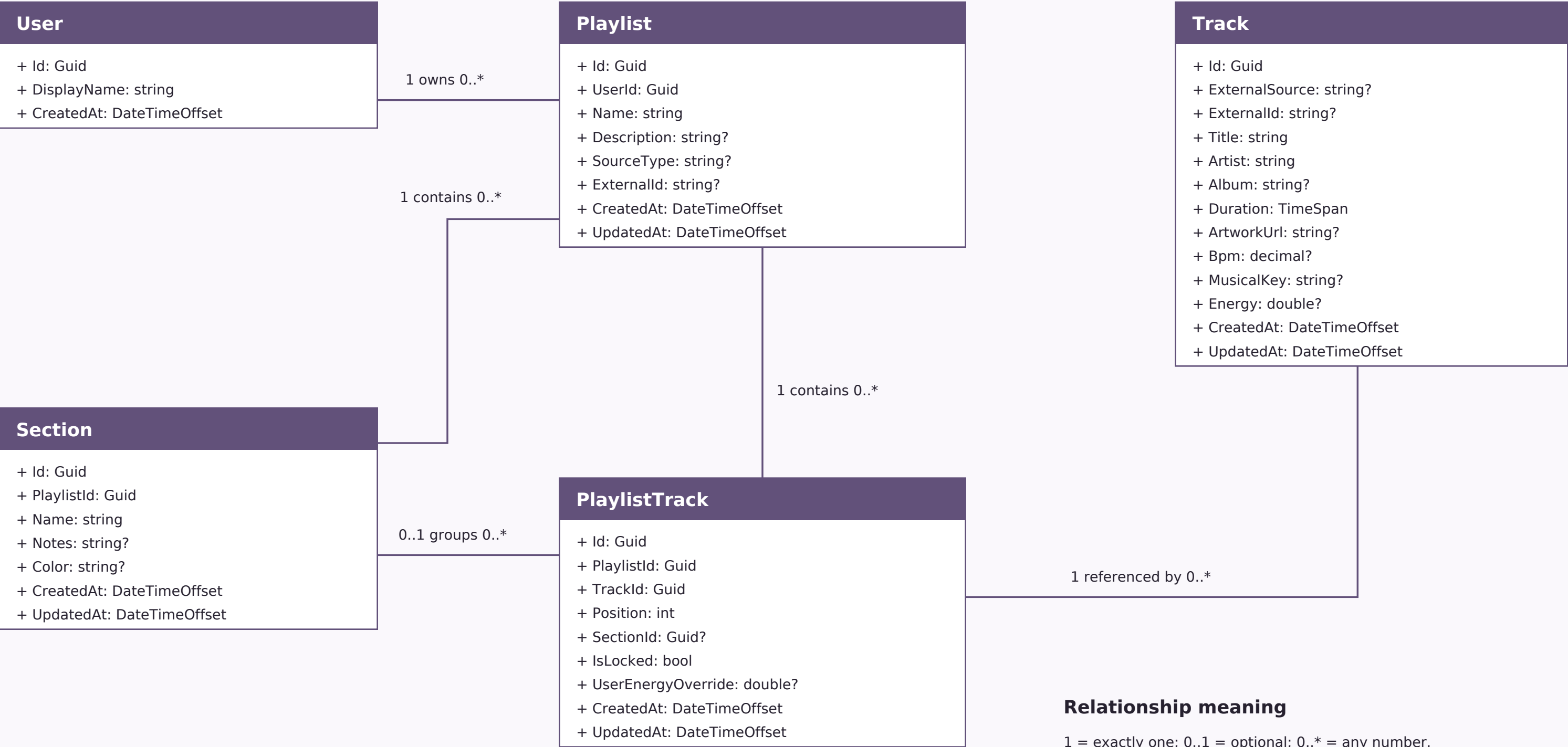


Core class diagram

IMPLEMENTED: five domain classes. Navigation properties are represented by associations.



Relationship meaning

1 = exactly one; 0..1 = optional; 0..* = any number.
PlaylistTrack represents one occurrence, not one song.
Repeated songs receive separate occurrence IDs.
Playback order comes from Position, not canvas location.
? = optional metadata or optional foreign key.
Required relationships remain required even when their navigation properties have not been loaded.

Versioning and transition plan

PLANNED: implement snapshots, draft save, and transition persistence together.



Additional associations and snapshot rules

Playlist 1 -> 0..* PlaylistVersion; User 1 -> 0..* PlaylistVersion (author); Track 1 -> 0..* PlaylistVersionTrack.

PlaylistVersion 1 -> 0..* TransitionNote. Both note endpoints belong to that version and must be adjacent occurrences.

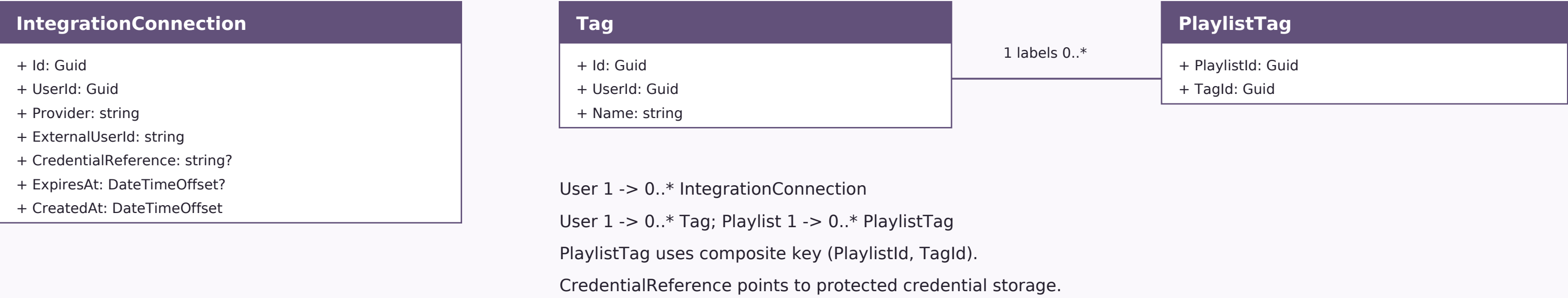
ParentVersionId is optional and references the same playlist. Add Playlist.CurrentVersionId with a same-playlist constraint.

Final versions are immutable. Snapshot sections and metadata preserve history; OriginalPlaylistTrackId is provenance, not a foreign key.

Transition endpoints use occurrence IDs so repeated songs are unambiguous. Calculated tempo/energy deltas are observations.

Integration, tagging, and persistence rules

PLANNED: provider adapters, credentials, and tags follow the core sequencing model.



Database and application constraints

Ownership	Authorize every playlist operation; tags and assigned sections must belong to the same owner/playlist.
Ordering	Unique (PlaylistId, Position); one-based contiguous positions. Reorder transactionally. Repeated TrackId is allowed.
Sections	Optional membership; each section forms a contiguous run. Removing a section unassigns its occurrences.
Metadata	Nonblank names/title/artist; duration >= 0; BPM > 0 when supplied; energy and overrides in [0, 1].
Energy	Effective value = occurrence override ?? shared track energy. Missing values stay null.
Deletion	Deleting playlist occurrences must not delete shared tracks. Preserve version history and configure cascade paths.
Versions	Unique version number per playlist and position per version; same-version note endpoints and section references.
Concurrency	Add optimistic concurrency and transactional snapshot save before persistence/autosave; update timestamps on writes.
Scope	Initial POCOs define structure only. Validation, locks, identity integration, EF Core, and migrations remain future work.

Editable Mermaid diagrams and detailed rules: docs/architecture/class-diagram.md